Teaching a Neural Network the Glow of the Solar System

A geometrically constrained model of zodiacal light for NASA's SPHEREx mission

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A geometrically constrained neural network learns the zodiacal light from viewing geometry alone, reaching ~0.03% median error on months it never saw.

Background

- **SPHEREx:** NASA space telescope, launched in March 2025
- Maps the **entire sky across a range of infrared wavelengths** (0.75–5 μm)
- **Science:** how the universe began, how galaxies formed, and where water and other ingredients for life are found

Zodiacal light

- Sunlight scattered by dust from comets and asteroids, plus the dust's heat glow (Fig. 1)
- Bright enough to be confused with faint light from distant galaxies
- Changes with look direction and Earth's place in orbit → **must be modeled and removed**

The Kelsall model

- Standard dust model (Kelsall et al. 1998, NASA's COBE): smooth cloud, three dust bands, a ring around the Sun, a clump trailing Earth
- **ZodiPy** computes what a telescope sees; we use it to simulate SPHEREx data

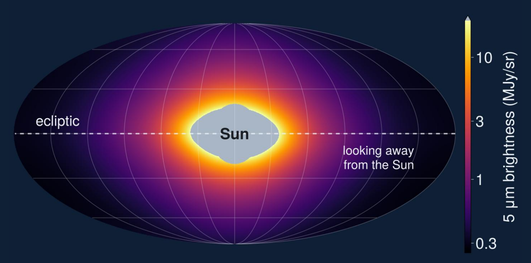


Figure 1. The whole sky at 5 μm from SPHEREx (ZodiPy): brightest toward the Sun and along the plane of the planets. Grey: too close to the Sun to observe.

The problem

- Zodiacal light must be removed from every SPHEREx image
- Physics models such as Kelsall's were fit to older data and leave errors behind

Research objective

- Build a **geometrically constrained** neural network that learns zodiacal light from SPHEREx data and **outperforms the Kelsall model**
- First step (this poster): learn ZodiPy's simulated sky from geometry alone

Methods

A geometrically constrained model

- No dates and no sky-map positions as inputs
- Only **9 numbers describing the viewing geometry** (Fig. 2): dot products of the look direction n, plus Earth's place in its orbit
- Must learn how brightness depends on geometry, as the physics does, not memorize when and where SPHEREx looked

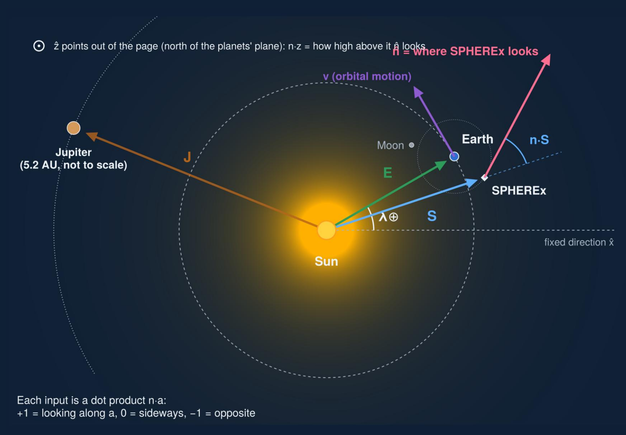


Figure 2. The geometry the network sees (top view, not to scale): n = where SPHEREx looks; S, E = Sun → SPHEREx, Earth; v = orbital motion; λ⊕ = Earth's angle around its orbit. Jupiter and the Moon: context only.

The 9 inputs

n·S, n·E	toward or away from the Sun (seen from SPHEREx and from Earth)
n·z	how far above or below the plane of the planets
n·vS, n·vE	ahead of or behind the orbital motion (SPHEREx, Earth)
n·x⊕, n·y⊕	look direction in a frame that turns with Earth (x⊕ = Sun → Earth)
sin λ⊕, cos λ⊕	Earth's position in its orbit: the time of year

Training and testing

- 423,607 simulated SPHEREx pointings from ZodiPy (140 days, 5 μm)
- Five small networks (4 hidden layers, 64 → 16 neurons; Flux.jl in Julia), averaged
- Tested on whole months and sky regions they never saw
- **Lesson:** adding the time of year (sin λ⊕, cos λ⊕) cut the errors ~40–65×
- **No overfitting:** error on withheld data levels off and never rises (Fig. 3)



Figure 3. Error during training on the training data and the withheld months: 5 networks (thin) and their average (thick). Test error ends ~2× the training error; with a random 10% withheld, the curves overlap.

Results

0.026% median error on months withheld from training

95% of pointings within 0.1% of ZodiPy

≤0.04% median error for every choice of withheld months

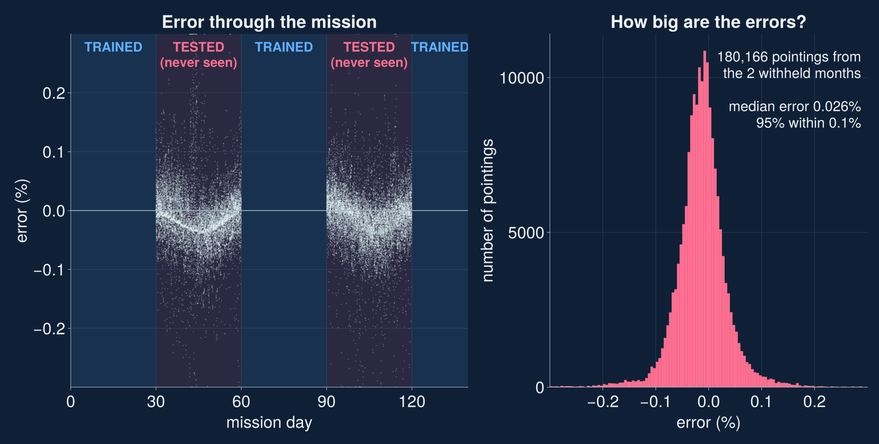


Figure 4. Network vs. ZodiPy on the two 30-day blocks never used in training (days 30–60 and 90–120; 180,166 pointings). Half of the errors are below 0.026%, 95% below 0.1%.

- **Robust:** six versions with different withheld months all reach 0.023–0.037% (Fig. 5)
- **Harder:** predicting past the start or end of the data (orange): fine for ~10 days, then errors grow

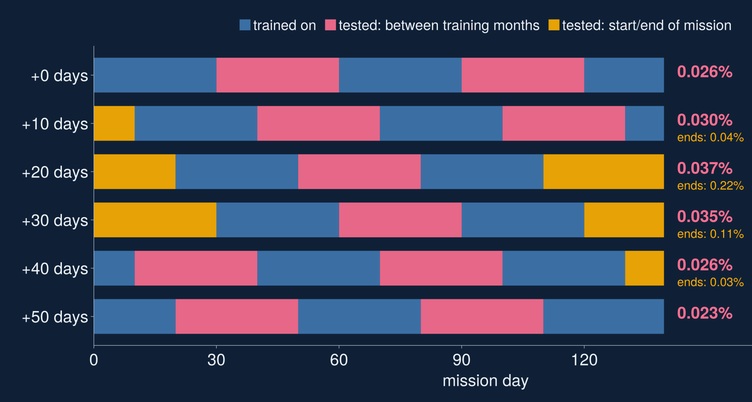


Figure 5. Six versions of the test, sliding the withheld months 10 days at a time. Right: median error on the red days (orange: start/end of the mission).

- **Largest errors:** along the plane of the planets, where the dust bands are (0.09%; Fig. 6)
- **Smallest:** toward the poles (0.016%)

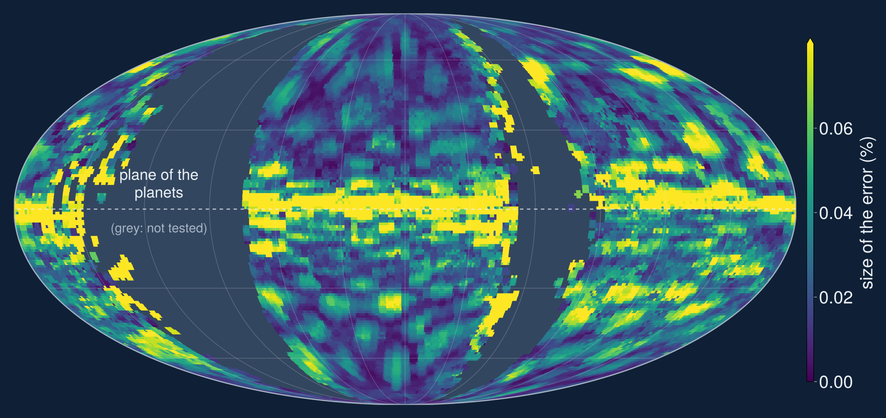


Figure 6. Error on pointings the network never saw, across the whole sky. Each pointing is drawn 3.5° wide, about SPHEREx's view. Grey: not tested.

Next Steps

✓ DONE Validate	Learn ZodiPy's simulated sky from geometry alone; test on unseen months and sky regions
IN PROGRESS Refine	Longer training; all SPHEREx wavelengths (a first 1.25–5 μm version: ~0.1% error)
NEXT Scale up	Move training to a computing cluster
NEXT Real data	Train on real SPHEREx images and compare with the Kelsall model

All results so far are preliminary and use simulated data.

Acknowledgements

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References

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